

Stationary Process : Its distribution remains unaffected by a time shift. $x(t)$ & $x(t+z)$ will have same dist

$$F_x(t_1, \dots, t_n) = F_x(t_1 + z, \dots, t_n + z)$$

$$f_x(x_1, x_2, \dots, x_n, t_1, t_2, \dots, t_n) = f(x_1, \dots, x_n, t_1 + z, \dots, t_n + z)$$

$$F_x(x, t) = F(x, t+z) = F_x(x)$$

$$\mu_x(t) = \mu \quad \text{Var}[x(t)] = \sigma^2$$

Wide Sense Stationary

If stationary condition is true only for $n \leq K$ where $K=2$; or its is stationary to order 2 then it is WSS.

① mean const

② $R_x(t, s) = R_x(|s-t|)$

Ergodic Process $\langle x(t) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$

$$\bar{R}_x(t) = \langle x(t), x(t+z) \rangle_{-T/2}^{T/2}$$

PAGE NO.
DATE: / /

$$= \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x(t) x(t+z) dt$$

A RP is ergodic if the time averages of sample funcⁿ of proc are equal to statistical mean / averages.

RP : A RV func of time

CDF : $F_x(x_1, t_1) = P\{x(t_1) \leq x_1\}$

PDF $f_x(x_1, t_1) = \frac{dF_x(x_1, t_1)}{dx_1}$

Independent RP : $f_x(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n)$
 $= f_x(x_1, t_1) \cdot f_x(x_2, t_2) \cdot \dots \cdot f_x(x_n, t_n)$

Stationary Random P / Strict sense
 if it is not affected by a shift in time

1st order : no change with shift in time
 $f_x(x, t_1) = f_x(x, t_1 + z)$
 mean constant

Proof : $x_1 = x_2(t_1) \quad x_2 = x_2(t_2)$

$E[x_1] = E[x(t_1)] = \int_{-\infty}^{\infty} x_1 f_x(x_1, t_1) dx_1$
 $t_2 = t_1 + z$

$E[x(t_2)] = \int_{-\infty}^{\infty} x_2 f_x(x_2, t_2) dx_2$

$$x_1 = x_2 = x$$

PAGE NO.

DATE

$$E[x(t_1 + z)] = \int_{-\infty}^{\infty} x_1 f_x(x_1, t_1 + z) dx_1$$

$$E[x(t_1)] = \int_{-\infty}^{\infty} x_1 f_x(x_1, t_1) dx_1$$

Proved

Second Order Stationary RP.

$$f_x(x_1, x_2; t_1, t_2) = f_x(x_1, x_2; t_1 + z, t_2 + z)$$

Condⁿ: Its autocoreⁿ funcⁿ should only depend on time difference not abs. time

$$R_{xx}(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$\text{If } \tau = t_2 - t_1, R_{xx}(t_1, t_1 + \tau) = E[x(t_1)x(t_1 + \tau)] = R_{xx}(\tau)$$

WSS: Second Order Stat process

Condⁿ: ① $\bar{x}$ const. ② $R_{xx}(\tau)$ on diff of time

Correlation funcⁿ: measure of similarity between the variables

2 types — ① Auto ② Cross

Auto core: measure of similarity b/w a given signal and replica of same signal by delayed time

$$t_1 = t \quad t_2 = t_1 + z \quad z = t_2 - t_1$$

$$R_{xx}(t_1, t_2) = E[x(t_1)x(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_x(x_1, x_2, t_1, t_2) dx_1 dx_2$$

$$R_{xx}(t_1, t_1 + z) = E[x(t_1)x(t_1 + z)]$$

Prop. of ACF : (1) mean sq value of RP obtained from R_{xx} by putting $z=0$
 $R_{xx}(0) = \overline{x^2} = E[x^2(t)]$

(2) $R_{xx}(z)$ is an even funcⁿ of z
 $R_{xx}(z) = R_{xx}(-z)$

$$R_{xx}(t, t+z) = R_{xx}(z) = E[x(t)x(t+z)]$$

$$z = -z \quad R_{xx}(-z) = E[x(t)x(t-z)]$$

$$t - z = a \quad t = a + z$$

$$R_{xx}(-z) = E[x(a+z)x(a)] = E_{xx}(z) = R_{xx}(z)$$

(3) $R_{xx}(z)$ at origin $z=0$ will have max value. All other values will be less than or equal than that $R_{xx}(0) \geq |R_{xx}(z)|$

$$R_{xx}(t_1, t_2) = E[x(t_1)x(t_2)]$$

consider $E\{[x(t_1)x(t_2)]\}^2 \geq 0$

$$E \left\{ x^2(t_1) + x^2(t_2) \pm 2x(t_1)x(t_2) \right\} \geq 0$$

$$E \left[x^2(t_1) \right] + E \left[x^2(t_2) \right] \pm 2E \left[x(t_1)x(t_2) \right] \geq 0$$

$$R_{xx}(0) + R_{xx}(0) \pm 2E \left[x(t_1)x(t_2) \right] \geq 0$$

$$R_{xx}(0) \geq |R_{xx}(z)|$$

$$(4) \quad z(t) = x(t) + y(t)$$

$$R_{(x+y)(x+y)}(z) = R_{xx}(z) + R_{xy}(z) + R_{yx}(z) + R_{yy}(z)$$

$$\begin{aligned} R_{zz}(z) &= R_{(x+y)(x+y)}(z) \\ &= E \left[z(t) z(t+z) \right] \end{aligned}$$

$$R_{zz}(z) = E \left[\{x(t) + y(t)\} \{x(t+z) + y(t+z)\} \right]$$

$$= E \left[x(t)x(t+z) \right] + E \left[x(t)y(t+z) \right]$$

$$= R_{xx}(z) + R_{xy}(z) + R_{yx}(z) + R_{yy}(z)$$

⑤ $x(t)$ has no periodic comp. $E[x(t)] \neq 0$
 $\lim_{z \rightarrow 0} R_{xx}(z) = \bar{x}^2$

$$R_{xx}(z) = E[x(t_1) x(t_1 + z)]$$

$x(t_1)$ & $x(t_2)$ are considered independent.

$$\lim_{z \rightarrow \infty} R_{xx}(z) = \lim_{z \rightarrow \infty} E[x(t_1) x(t_2)]$$

$$= E[x(t_1)] E[x(t_2)] = \bar{x}^2$$

⑥ no periodic comp. and $\bar{x} = 0$
 $\lim_{z \rightarrow \infty} R_{xx}(z) = 0$

⑦ periodic comp, then $R_{xx}(z)$ will also have periodic comp. with same period $R_{xx}(z \pm T) = R_{xx}(z)$

$$R_{xx}(z) = E[x(t) x(t + z)]$$

$$R_{xx}(z \pm T) = E[x(t) x(t + z \pm T)]$$

$$x(t + z \pm T) = x(t + z)$$

$$R_{xx}(z \pm T) = E[x(t) x(t + z)]$$

$$= R_{xx}(z)$$

8. If $x(t)$ has DC comp then R_{xx} will also have DC comp.
 $R_{xx}(z) = k^2 + R_{yy}(z)$

$$x(t) = k + y(t)$$

$$R_{xx}(z) = E[x(t)x(t+z)]$$

$$= E[(k + y(t))(k + y(t+z))]$$

$$= E[k^2]$$

$$y(t) = \begin{matrix} k^2 + kE[y(t+z)] + kE[y(t)] + \\ E[y(t)]E[y(t+z)] \end{matrix}$$

$$R_{xx}(z) = k^2 + R_{yy}(z)$$

9. $R_{xx}(z)$ of $x(t)$ is a finite energy func.

$$R_{xx}(t_1, t_2) = \int_{-\infty}^{\infty} R_{xx}(z) dz < \infty$$

10. $R_{xx}(z)$ cant have an arbitrary shape

Cross Correlation funcⁿ

Similarity of a signal and a time delayed signal of another funct.

$$R_{xy}(t_1, t_2) = E[x(t_1) y(t_2)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{xy}(x, y, t_1, t_2) dx dy$$

$$t_1 = t \quad t_2 = t_1 + z \quad z = t + z$$

$$R_{xy}(t, t+z) = E[x(t) y(t+z)]$$

• If $y_1 = y(t) \quad x_2 = x(t+z)$

$$R_{yx}(z) = E[y(t) x(t+z)]$$

• If $R_{yx}(t, t+z) = 0 \quad x(t) \& y(t)$ are orthogonal process

• $x(t) \& y(t)$ independent
 $= E[x(t)] E[y(t+z)]$

• Indp + WSS $R_{xx}(z) = \overline{xy} = \text{const}$

Prop of CCF:

① $R_{xy}(z) = R_{yx}(-z)$ is a symm prop

$$R_{xy}(z) = E[x(t) y(t+z)]$$

$$R_{yx}(z) = E[y(t) x(t+z)]$$

$$z = -z \quad R_{yx}(-z) = E[y(t) x(t-z)]$$

$$t - z = u \quad t = u + z$$

$$R_{yz}(-z) = E[Y(u+z)X(u)] = R_{xy}(z)$$

2. $x(t)$ & $y(t)$ $R_{xx}(z)$ $R_{yy}(z)$

$$|R_{xy}(z)| \leq \sqrt{R_{xx}(0)R_{yy}(0)}$$

$$E \left[\frac{x(t)}{\sqrt{R_{xx}(0)}} \pm \frac{y(t)}{\sqrt{R_{yy}(0)}} \right]^2 \geq 0$$

$$E \left\{ \frac{x^2(t)}{R_{xx}(0)} + \frac{y^2(t)}{R_{yy}(0)} \pm \frac{2x(t)y(t)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \right\} \geq 0$$

$$\frac{E[x^2(t)]}{R_{xx}(0)} + \frac{E[y^2(t)]}{R_{yy}(0)} \pm \frac{2E[x(t)y(t)]}{\sqrt{R_{xx}(0)R_{yy}(0)}}$$

$$2 \pm \frac{2R_{xy}(z)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \geq 0$$

$$2 \geq \pm \frac{2R_{xy}(z)}{\sqrt{R_{xx}(0)R_{yy}(0)}}$$

$$|R_{xy}(z)| \leq \sqrt{R_{xx}(0)R_{yy}(0)}$$

3. $|R_{xy}(z)| \leq \frac{R_{xx}(0) + R_{yy}(0)}{2}$

$$AM \leq GM \quad \sqrt{R_{xx}(0)R_{yy}(0)} \leq \frac{R_{xx}(0) + R_{yy}(0)}{2}$$

$$|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)R_{yy}(0)}$$

$$|R_{xy}(\tau)| \leq \frac{R_{xx}(0) + R_{yy}(0)}{2}$$

⑤ $x(t)$ & $y(t)$ indep $\Rightarrow$

$$R_{xy}(\tau) = E[x(t)] E[y(t)]$$

$$E[y(t+\tau)] = E[y(t)]$$

$$R_{xy}(\tau) = E[x(t)] E[y(t)]$$

⑥ mean $x(t)$ & $y(t) = 0$
or WSS

$$\lim_{\tau \rightarrow \infty} R_{xy}(\tau) = 0$$

$$\lim_{\tau \rightarrow \infty} R_{xy}(\tau) = \lim_{\tau \rightarrow \infty} E[x(t) y(t+\tau)]$$

$$= \lim_{\tau \rightarrow \infty} E[x(t)] E[y(t+\tau)]$$

$$= 0$$

⑦ $x(t)$ & $y(t)$ are periodic (T_0).
 $R_{xy}(\tau)$ will have periodic comp with T_0 .

⑧ uncorrelated if $R_{xy}(\tau) = E[x(t)] E[y(t)]$

PAGE NO. _____
DATE: ____/____/____

Ergodic time averages of $x(t)$ are equal to corresponding statistical average of $x(t)$.

$$E[x(t)] = \langle x(t) \rangle \quad R_{xx}(z) = \langle R_{xx}(z) \rangle$$

Jointly Ergo: Individually ergodic & ~~to~~ cross cross funcⁿ are equal to resp CCF

$$R_{xx}(z) = \langle R_{xx}(z) \rangle$$

$$R_{xy}(z) = \langle R_{yx}(z) \rangle$$

• Mean Ergodic Process: time average of sample func $x(t)$ equal to $\bar{x}$.

$$E[x(t)] = \bar{x} = A[x(t)] = \bar{x}$$

• Auto cross Ergo P: $A[x(t)x(t+z)] = E[x(t)x(t+z)]$
 $\langle R_{xx}(z) \rangle = R_{xx}(z)$

• Cross cross CEP: $A[x(t)y(t+z)] = \langle R_{xy}(z) \rangle = R_{xy}(z)$
 $E[x(t)y(t+z)]$